

The big picture

Guiding question(s)

How does the application of the law of conservation of energy help us to predict energy changes during reactions?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

The moment food enters our body it begins to undergo a remarkable transformation. Chemical reactions alter carbohydrates, fats, and proteins into a vast range of molecules that our body uses in a myriad of processes. One important metabolic product is glucose, $C_6H_{12}O_6$, and this molecule can be thought of as chemical energy, used to fuel our body. But we do not simply turn our dinner into energy to go for a run. Instead, a series of chemical steps ultimately results in glucose and the subsequent reactions of cellular respiration.

This is a working example of the law of conservation of energy: food energy is not lost, it is converted into forms of energy to fuel our body, allowing us to interact with the world around us. How does this knowledge help us to make informed food choices for our activities?

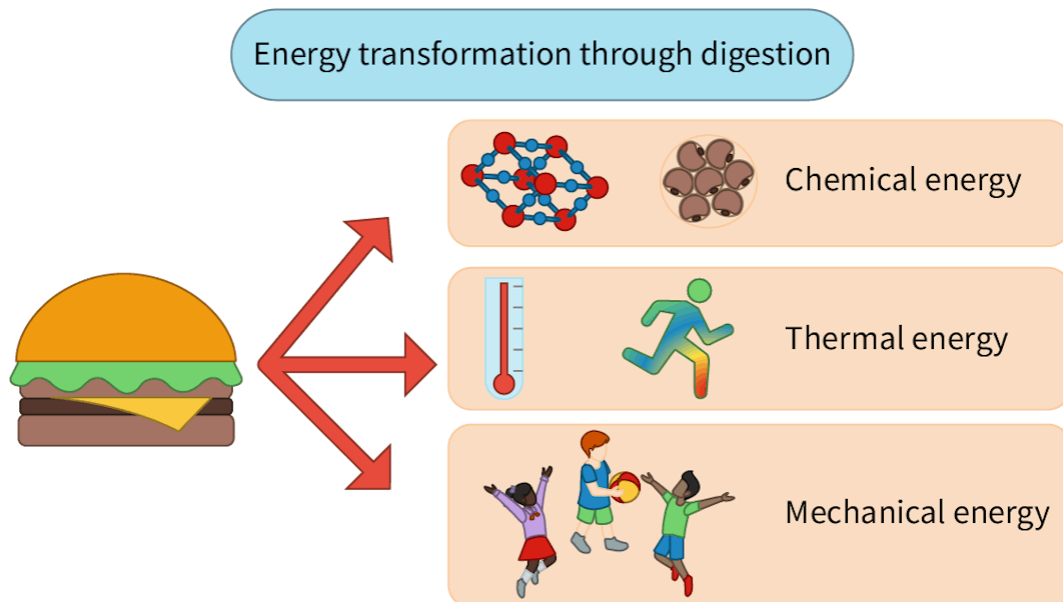


Figure 1. Food is transformed into chemical, thermal, and mechanical energy.

👁 More information for figure 1

Energy use in our body is a complicated process and involves reactions that cannot be measured individually. Energy cycles allow us to calculate the energy in a reaction through alternate pathways when we may not be able to measure it directly. The use of energy cycles in reactions provides the framework to do so, employing a variety of strategies that are examined in this section. How does this all relate to the law of conservation of energy?

Concept

Aspect: Reactivity

The law of conservation of energy states that energy can neither be created nor destroyed in a closed system which is isolated from the surroundings but is instead converted from one form to another.

Prior learning

Before you study this subtopic make sure that you understand the following:

- That ionic compounds exist as three-dimensional lattice structures, represented by empirical formulas (see [section S2.1.3a](#)  (</study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/>)).
- That reactions are described as endothermic or exothermic, depending on the direction of energy transfer between the system and the surroundings (see [section R1.1.2](#)  (</study/app/preview-p/sid-426-cid-753301/book/endothermic-and-exothermic-reactions-id-45282/>)).
- That the relative stability of reactants and products determines whether reactions are endothermic or exothermic (see [section R1.1.3](#)  (:sectionlink:138726)).